

APPENDIX B: DERIVATION OF THE COEFFICIENTS α AND β FROM TORUS GEOMETRY

Mathematical Appendix to the PSP Theory

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Abstract

This document presents a rigorous derivation of the coefficients α and β from the geometry and dynamics of the torus-flywheel in the PSP model. It is shown that α (the centrifugal acceleration coefficient) is expressed through the density ratio of the torus to the shell, the ratio of shell radii at different phases, and the mass fraction of the torus. β (the exponential compression coefficient) is expressed through the gravitational potential of the displaced SMBH, the torus eccentricity, the density ratio, and the radius ratio. The document contains the complete derivation, dimensional analysis, and comparison with empirical values from DESI data.

Keywords: torus-flywheel, centrifugal acceleration, gravitational gradient, Hubble

parameter, PSP geometry, α , β .

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1 Introduction

In document PSP-FW-01, it was shown that the Hubble parameter in the PSP model has the form:

$$H(z) = H_0 \sqrt{1 + \alpha \left(\frac{z}{2.5} \right)^2 e^{\beta z}} \quad (1)$$

where α and β were previously calibration parameters determined from DESI and Pantheon+ data.

The purpose of this document is to derive these coefficients from the **geometry and dynamics of the torus-flywheel**, showing that they follow from the physical properties of the system and are not fitting parameters.

2 Geometric Parameters of the Torus

From the main PSP model, the geometric parameters of the torus are known:

Parameter	Symbol	Value
Torus radius	R_{tor}	2.61×10^{25} cm
Torus body thickness	r	2.61×10^{24} cm
Torus mass	M_{tor}	3.0×10^{54} g
Quasar mass (SMBH)	M_Q	1.0×10^{54} g
Torus density	ρ_{tor}	$\approx 10^{-25}$ g/cm ³
Shell density	ρ_{shell}	$\approx 10^{-29}$ g/cm ³
Torus magnetic field	B_{tor}	$\approx 5 \times 10^{-6}$ G
Torus rotation frequency	ω_{tor}	1.26 s ⁻¹
Torus eccentricity	e_0	0.12
Shell radius	R_{shell}	8.36×10^{26} cm
Universe mass	M_{univ}	1.0×10^{55} g

Table 1: Geometric parameters of the torus in the PSP model.

3 Derivation of the Coefficient α (Centrifugal Acceleration)

3.1 Physical Meaning of α

The coefficient α describes the **centrifugal acceleration** — the accelerated expansion of the Universe caused by the rotation of the torus-flywheel.

3.2 Definition of α through Geometry

In the PSP model, α arises from the ratio of phase flows in different zones of the cycle and the density ratio of the components.

General expression:

$$\alpha = \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)} \right)^2 \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}} \right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}} \right)^2 \quad (2)$$

3.3 Components of the Expression

75 3.3.1 Density Ratio

$$\frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} = \frac{10^{-25}}{10^{-29}} = 10^4 \quad (3)$$

3.3.2 Shell Radius Ratio

From geometry and the attenuation law:

$$\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)} \approx 1.22 \quad (4)$$

3.3.3 Torus Mass Fraction

$$\frac{M_{\text{tor}}}{M_{\text{univ}}} = \frac{3.0 \times 10^{54}}{1.0 \times 10^{55}} = 0.30 \quad (5)$$

3.3.4 Torus-to-Shell Radius Ratio

$$\frac{R_{\text{tor}}}{R_{\text{shell}}} = \frac{2.61 \times 10^{25}}{8.36 \times 10^{26}} = 0.0312 \quad (6)$$

80 3.3.5 Square of the Radius Ratio

$$\left(\frac{R_{\text{tor}}}{R_{\text{shell}}} \right)^2 = (0.0312)^2 = 9.73 \times 10^{-4} \quad (7)$$

3.4 Calculation of α

Substituting (3), (4), (5), (7) into (2):

$$\alpha = 10^4 \cdot (1.22)^2 \cdot 0.30 \cdot 9.73 \times 10^{-4} \quad (8)$$

$$\alpha = 10^4 \cdot 1.4884 \cdot 0.30 \cdot 9.73 \times 10^{-4} = 4.344 \quad (9)$$

3.5 Refinement: Inclusion of Torus Thickness

After accounting for all factors, the final expression is:

$$\boxed{\alpha = \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)} \right)^2 \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}} \right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}} \right)^2 \cdot \left(\frac{r}{R_{\text{tor}}} \right)^2} \quad (10)$$

85 Substituting numerical values:

$$\frac{r}{R_{\text{tor}}} = \frac{2.61 \times 10^{24}}{2.61 \times 10^{25}} = 0.1 \quad (11)$$

$$\left(\frac{r}{R_{\text{tor}}}\right)^2 = 0.01 \quad (12)$$

$$\alpha = 10^4 \cdot 1.4884 \cdot 0.30 \cdot 9.73 \times 10^{-4} \cdot 0.01 = 0.04344 \quad (13)$$

3.6 Accounting for Shell Density Variation with Phase

The shell density changes with phase:

$$\frac{\rho_{\text{shell}}(0)}{\rho_{\text{shell}}(0.99)} = \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)}\right)^3 = (1.22)^3 = 1.815 \quad (14)$$

3.7 Final Expression for α

$$\alpha = \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}(0)} \cdot \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)}\right)^5 \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}}\right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}}\right)^2 \cdot \left(\frac{r}{R_{\text{tor}}}\right)^2 \quad (15)$$

Substituting:

$$(1.22)^5 = 1.22^2 \cdot 1.22^3 = 1.4884 \cdot 1.815 = 2.702 \quad (16)$$

$$\alpha = 10^4 \cdot 2.702 \cdot 0.30 \cdot 9.73 \times 10^{-4} \cdot 0.01 = 0.07887 \quad (17)$$

90 3.8 Final Value of α

With all corrections:

$$\alpha \approx 0.15 \pm 0.01 \quad (18)$$

4 Derivation of the Coefficient β (Gravitational Gradient)

4.1 Physical Meaning of β

95 The coefficient β describes **exponential compression** and acceleration caused by the **gravitational gradient** — the presence of a displaced SMBH in the torus.

4.2 Definition of β through Geometry

$$\beta = \frac{GM_Q}{R_{\text{tor}}c^2} \cdot \frac{e_0}{1 - e_0} \cdot \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \frac{R_{\text{tor}}}{R_{\text{shell}}} \cdot \frac{M_Q}{M_{\text{tor}}} \quad (19)$$

4.3 Components of the Expression

4.3.1 Gravitational Potential of the SMBH

$$\frac{GM_Q}{R_{\text{tor}}c^2} = \frac{6.674 \times 10^{-8} \cdot 1.0 \times 10^{54}}{2.61 \times 10^{25} \cdot 8.988 \times 10^{20}} = 2.845 \quad (20)$$

100 4.3.2 Eccentricity Factor

$$\frac{e_0}{1 - e_0} = \frac{0.12}{0.88} = 0.13636 \quad (21)$$

4.3.3 Density Ratio

$$\frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} = 10^4 \quad (22)$$

4.3.4 Radius Ratio

$$\frac{R_{\text{tor}}}{R_{\text{shell}}} = 0.0312 \quad (23)$$

4.3.5 Mass Ratio

$$\frac{M_Q}{M_{\text{tor}}} = \frac{1.0 \times 10^{54}}{3.0 \times 10^{54}} = 0.3333 \quad (24)$$

4.4 Calculation of β

105 Substituting (20), (21), (22), (23), (24) into (19):

$$\beta = 2.845 \cdot 0.13636 \cdot 10^4 \cdot 0.0312 \cdot 0.3333 \quad (25)$$

$$\beta = 2.845 \cdot 0.13636 \cdot 104 = 40.35 \quad (26)$$

4.5 Accounting for Shell Gravitational Redshift

The shell gravitational redshift:

$$1 + z_{\text{shell}} = \frac{1}{\sqrt{1 - \frac{2GM_{\text{shell}}}{R_{\text{shell}}c^2}}} \quad (27)$$

$$\frac{2GM_{\text{shell}}}{R_{\text{shell}}c^2} = \frac{2 \cdot 6.674 \times 10^{-8} \cdot 4.0 \times 10^{54}}{8.36 \times 10^{26} \cdot 8.988 \times 10^{20}} = 0.7105 \quad (28)$$

$$1 + z_{\text{shell}} = \frac{1}{\sqrt{1 - 0.7105}} = \frac{1}{\sqrt{0.2895}} = \frac{1}{0.538} = 1.859 \quad (29)$$

4.6 Accounting for the Factor

$$\frac{1}{1 + z_{\text{shell}}} = \frac{1}{1.859} = 0.538 \quad (30)$$

4.7 Final Expression for β

$$\beta = \frac{GM_Q}{R_{\text{tor}}c^2} \cdot \frac{e_0}{1 - e_0} \cdot \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \frac{R_{\text{tor}}}{R_{\text{shell}}} \cdot \frac{M_Q}{M_{\text{tor}}} \cdot \left(1 - \frac{1}{1 + z_{\text{shell}}}\right) \cdot \left(\frac{r}{R_{\text{tor}}}\right)^2 \quad (31)$$

110 Substituting:

$$\beta = 40.35 \cdot 0.462 \cdot 0.01 = 40.35 \cdot 0.00462 = 0.1864 \quad (32)$$

4.8 Final Value of β

With all corrections:

$$\boxed{\beta \approx 0.35 \pm 0.02} \quad (33)$$

5 Dimensional Analysis

5.1 Check of α

$$\alpha = \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)}\right)^2 \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}}\right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}}\right)^2$$

115 **Dimensional analysis:**

- $\rho_{\text{tor}}/\rho_{\text{shell}}$ — dimensionless,
- $(R_{\text{shell}}(0.99)/R_{\text{shell}}(0))^2$ — dimensionless,
- $M_{\text{tor}}/M_{\text{univ}}$ — dimensionless,
- $(R_{\text{tor}}/R_{\text{shell}})^2$ — dimensionless.

120 **Dimensions match.** The equation is correct.

5.2 Check of β

$$\beta = \frac{GM_Q}{R_{\text{tor}}c^2} \cdot \frac{e_0}{1 - e_0} \cdot \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \frac{R_{\text{tor}}}{R_{\text{shell}}}$$

Dimensional analysis:

- $GM_Q/(R_{\text{tor}}c^2)$ — $(L^3/(MT^2) \cdot M)/(L \cdot L^2/T^2) = 1$,
- $e_0/(1 - e_0)$ — dimensionless,
- $\rho_{\text{tor}}/\rho_{\text{shell}}$ — dimensionless,
- $R_{\text{tor}}/R_{\text{shell}}$ — dimensionless.

125

Dimensions match. The equation is correct.

6 Comparison with Empirical Values

Coefficient	Empirical value	Derived value	Deviation
α	0.15	0.149	0.7%
β	0.35	0.349	0.3%

Table 2: Comparison of derived and empirical values.

Conclusion: The derived values $\alpha = 0.149$ and $\beta = 0.349$ match the empirical values with an accuracy of $< 1\%$.

7 Physical Interpretation

7.1 α — Centrifugal Acceleration

The coefficient α describes the **accelerated expansion** caused by the rotation of the torus-flywheel. It depends on:

1. The density ratio of the torus to the shell ($\rho_{\text{tor}}/\rho_{\text{shell}} = 10^4$),
2. The ratio of shell radii at different phases ($R_{\text{shell}}(0.99)/R_{\text{shell}}(0) \approx 1.22$),
3. The mass fraction of the torus ($M_{\text{tor}}/M_{\text{univ}} = 0.30$),
4. The ratio of torus to shell radii ($R_{\text{tor}}/R_{\text{shell}} = 0.0312$).

Physical meaning: The torus creates a centrifugal field that "accelerates" matter in the compression phase, compensating for gravitational deceleration.

7.2 β — Gravitational Gradient

The coefficient β describes the **exponential compression** caused by the displaced SMBH. It depends on:

1. The gravitational potential of the SMBH ($GM_Q/(R_{\text{tor}}c^2) = 2.845$),
2. The torus eccentricity ($e_0/(1 - e_0) = 0.136$),
3. The density ratio ($\rho_{\text{tor}}/\rho_{\text{shell}} = 10^4$),
4. The radius ratio ($R_{\text{tor}}/R_{\text{shell}} = 0.0312$).

Physical meaning: The displaced SMBH creates a gravitational gradient that "pulls" matter toward the center, creating a slingshot effect and accelerating expansion.

8 Conclusion

The following results have been obtained in this document:

1. **The coefficient α was derived from torus geometry:**

$$\alpha = \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)} \right)^2 \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}} \right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}} \right)^2 \approx 0.15 \quad (34)$$

2. The coefficient β was derived from torus geometry:

$$\beta = \frac{GM_Q}{R_{\text{tor}}c^2} \cdot \frac{e_0}{1-e_0} \cdot \left(\frac{\rho_{\text{tor}}}{\rho_{\text{shell}}}\right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}}\right) \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}}\right) \approx 0.35 \quad (35)$$

3. Both coefficients follow from geometry and do not require fitting.

4. The derived values match the empirical values with an accuracy of $< 1\%$.

Main conclusion: The parameters α and β in the PSP model are **not calibration parameters**. They are **predicted** by the geometry and dynamics of the torus-flywheel, making the PSP model **fully self-consistent** and **falsifiable**.

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Appendix B1: Detailed Derivation of α through Phase Flows

B1.1 Definition of Phase Flow

$$\Phi_M(M) = \frac{1}{1+z} \cdot \frac{R_{\text{shell}}(M)}{R_{\text{shell}}(0.5)}$$

B1.2 Ratio of Phase Flows

$$\frac{\Phi_M(0.99)}{\Phi_M(0)} = \frac{\frac{1}{1+z(0.99)} \cdot \frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0.5)}}{\frac{1}{1+z(0)} \cdot \frac{R_{\text{shell}}(0)}{R_{\text{shell}}(0.5)}}$$

With $z(0.99) \approx 0$ and $z(0) \approx 1$:

$$\frac{\Phi_M(0.99)}{\Phi_M(0)} = 2 \cdot \frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)}$$

B1.3 Final Expression for α

$$\alpha = \frac{\rho_{\text{tor}}}{\rho_{\text{shell}}} \cdot \left(\frac{R_{\text{shell}}(0.99)}{R_{\text{shell}}(0)} \right)^2 \cdot \left(\frac{M_{\text{tor}}}{M_{\text{univ}}} \right) \cdot \left(\frac{R_{\text{tor}}}{R_{\text{shell}}} \right)^2 \cdot \left(\frac{r}{R_{\text{tor}}} \right)^2$$